

- Attaching tooling, utility boxes, solenoid packs, vacuum pumps, cameras, lighting, etc., to the robot tool mount flange.
- Attaching hoses, pneumatic lines, or cables to the robot. These should be designed so they do not restrict joint motion or cause robot motion errors.

2.12.2 Unacceptable Modifications

The modifications listed below may damage the robot, reduce system safety and reliability, or shorten the life span of the robot. The warranty of the entire robot or certain parts may be voided.



CAUTION: Making any of the modifications outlined below voids the warranty of any components that the manufacturer determines were damaged due to the modification.

Contact your supplier if you are considering any of the following modifications:

- Modifying any of the robot harnesses cables
- Modifying any robot access covers or drive system components
- Modifying (including drilling or cutting) any robot surface
- Modifying any robot electrical component or printed-circuit board
- Routing additional hoses, air lines, or wires through the inside of the robot
- Making any modifications that compromise EMC performance, including shielding

2.12.3 Safety Requirements for Additional Equipment

Additional equipment used with the WHR (grippers, conveyor belts, etc.) must not reduce the overall work cell safeguards.

Emergency stop switches must be accessible at all times.

If the robot is to be used in an EU or EEA member country, all components in the work cell must comply with the safety requirements in the European Machine Directive 89/392/EEC (and subsequent amendments) and related harmonized European, international, and national standards. For robot systems, these include: EN 775/ISO 10218, sections 5, 6; EN 292-2; and EN 60204. For safety fences, see EN 294.

In other countries, it is strongly recommended that a similar level of safety, and compliance with applicable local and national regulations, including SEMI-S2, be maintained.

2.13 QUALIFICATION OF PERSONNEL

This manual assumes that all personnel have attended a training course and have a working knowledge of the system. The user must provide the necessary additional training for all personnel who will be working with the system.

As noted in this handbook, certain procedures should be performed only by skilled or instructed persons. The following standard terms are being used:

- Skilled persons have technical knowledge or sufficient experience to enable them to avoid relevant dangers, whether electrical and/or mechanical.
- Instructed persons are adequately advised or supervised by skilled persons to enable them to avoid relevant dangers, whether electrical or mechanical.

All personnel must observe industry-prescribed safety practices during the installation, operation, and testing of any electrically powered equipment. To avoid injury or damage to equipment, always remove power by disconnecting the AC power from the source before attempting any

repair or upgrade activity. Use appropriate lockout procedures to reduce the risk of power being restored by another person while an operator is working on the system.



WARNING: The user must receive confirmation from every entrusted individual before they do any work with the WHR that the person:

- Has received the User Manual;
- Has read the User Manual;
- Understands the User Manual;
- Will work in the manner specified by the User Manual.

2.14 SAFETY EQUIPMENT FOR OPERATORS

Operators must wear the following safety equipment while inside the device workcell:

- Protective eyewear aka safety glasses;
- Protective headgear (hard hats);
- Safety shoes.

Warning signs should be clearly posted around the workcell to ensure that anyone working around the system knows they must wear this safety equipment.

2.15 PROTECTION AGAINST UNAUTHORIZED OPERATION

The system must be protected against unauthorized use. The user or operator must restrict access to the keyboard and Teach Pendant by placing them in secure/locked storage, or through the use of another adequate method to prevent unauthorized access.

2.16 SAFETY ASPECTS WHILE PERFORMING MAINTENANCE

Only skilled persons with the necessary knowledge of the safety and operation of the equipment are allowed to perform maintenance on the robot, controller, and power chassis.



CAUTION: During maintenance and repair, the power of the WHR must be turned off. Lockout measures must be used to prevent unauthorized personnel from turning on the power.

2.17 WHAT TO DO IN AN EMERGENCY SITUATION

Press any E-Stop button (a red push-button on a yellow background/field) and then follow the appropriate internal emergency procedures of your company or organization. In case of fire, use CO₂ to extinguish.

2.18 ADDITIONAL SAFETY INFORMATION

The standards and regulations listed in this manual contain additional guidelines for wafer systems installation, safeguarding, maintenance, testing, start-up, and operator training. The following table lists some sources for various standards:

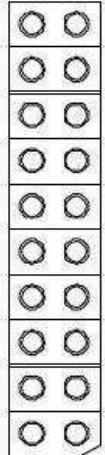
1	Ground	14	Ground
2	InputE	15	OutputE
3	InputF	16	OutputF
4	InputG	17	OutputG
5	InputH	18	OutputH
6	InputJ	19	OutputJ
7	InputK	20	OutputK
8	InputL	21	OutputL
9	InputM	22	OutputM
10	InputN	23	OutputN
11	InputR	24	Ground
12	PullUp/24VDC	25	PullUp/24VDC
13	Reserved		

NOTE: Pins 12 and 25 can provide a total of 0.2A current.

3.1.9 Sensor Type Configuration

WHR controller provides two jumper blocks for selecting of sensor types (PNP or NPN) connected to controller' input lines, as follows:

Jumper Block J7 – Arm Sensors Configuration (next to Arm Interface Connector, CN16)			
	Sensor D	ON: Pull Up/NPN (default)	Off: Pull Down/PNP
	Sensor C	ON: Pull Up/NPN (default)	Off: Pull Down/PNP
	Sensor B	ON: Pull Up/NPN (default)	Off: Pull Down/PNP
	Sensor A	ON: Pull Up/NPN (default)	Off: Pull Down/PNP
	Home R	ON: Pull Up/NPN (default)	Off: Pull Down/PNP

Jumper Block J10 – Extended I/O Configuration (next to Extended Interface Connector, CN23)			
	Input E	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input F	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input G	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input H	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input J	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input K	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input L	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input M	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input N	ON: Pull Up/NPN	Off: Pull Down/PNP (default)
	Input R	ON: Pull Up/NPN	Off: Pull Down/PNP (default)

3.2 END-EFFECTOR VACUUM LINES

The single-arm robot is delivered with a solenoid valve and internal vacuum sensor for a single vacuum end effector. Two 3mm quick connect fittings are located on the robots interface panel. For standard configurations use the top quick connect fitting. Insert a 3mm air line into the vacuum fitting on the connector panel of the robot.



CAUTION: Do not apply a positive pressure to either of the vacuum ports. This could cause damage to the robot.

A factory installed dual valve option is available. For the dual valve option the quick connect fittings are used. The upper fitting is dedicated to the first end-effector and the lower fitting is for the second end-effector.

The dual-arm robot is delivered as standard with dual solenoid valves and internal vacuum sensors, one per each paddle. Two 3mm quick connect fittings are located on the robot interface panel. For standard configurations use the top quick connect fitting. Insert a 3mm air line into the vacuum fitting on the connector panel of the robot.

Vacuum sensors' sensitivity can be adjusted by the following procedure:

- On dual-arm robots, lift the Z axis by about 5" to facilitate access to retaining screws.

MVA Z 5000

- Turn servo loop off:

SOF

- On single-arm robot: Remove the cover of the lower arm link by removing the screws that attach the cover to the arm.
- On dual-arm robot: Remove the covers on the octa-box by removing the screw attaching the covers from underneath.
- Connect vacuum line(s) and install end effector paddle(s)
- Open vacuum valve 1 or 2:

VON 0

or

VON 1

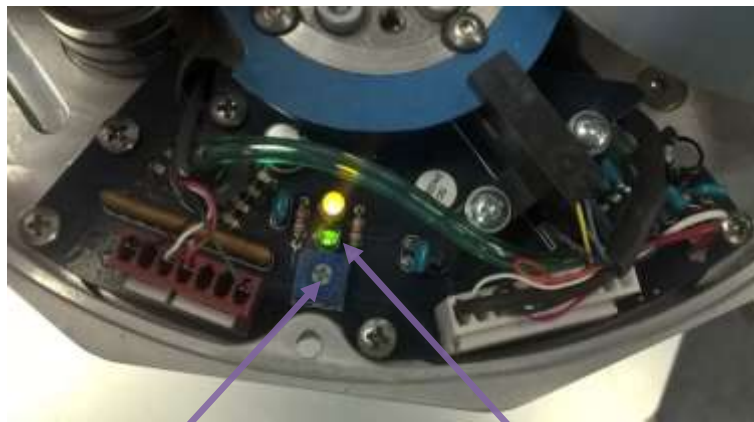
- With no wafer placed on the end effector, turn the potentiometer located on the vacuum sensor slowly counter-clockwise until the corresponding LED turns on. Mark this position as A.
- With a wafer placed on the end effector, turn the potentiometer clockwise until the corresponding LED turns off. Mark this position as B.
- Turn the potentiometer midway between positions A and B
- Repeat the procedure for the other end effector, if such is installed.
- Replace the cover(s).

By customer request, external flange-mounted vacuum sensors with indicators can be installed. Please refer to sensor' documentation for details about setting up sensitivity.



Potentiometers 1 and LEDs 1 and 2

Figure 3-3: Single-arm robot : Dual vacuum line control board in lower arm link



Potentiometer

Sensor LED

Figure 3-4: Dual-arm robot: Single vacuum line in octa-box, for each arm